Environment Separation

Ref: 2.3
Product: iConnect Plus
Organisation: Cache Digitech
Classification: Internal — Confidential
Version: 2.0
Date: 2026-11-03
Deployment model: On-premises (Cache Digitech premises) · VPN for remote access
Audience: Platform engineering, IT, security assessors

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1. Purpose

Describe how **development**, **test/UAT**, and **production** environments are separated on **Cache Digitech internal infrastructure**, including access paths (LAN/VPN) and privileged controls.

2. Environment summary

Environment	Purpose	Host location	Data classification
Development	Feature work, local Compose	Engineer machines / dev VM	Synthetic / anonymized preferred
Test / UAT	Integration, QA, stakeholder sign-off	Internal non-prod server or VLAN	Masked copy of prod — **Template / Pending sign-off**
Production	Live business operations	Cache Digitech premises (e.g. `172.16.200.30`)	Confidential / regulated

All environments run the **same stack pattern** (Docker Compose: infra + app); they differ by **host**, **credentials**, **data**, and **network ACLs**.

3. Separation controls

Control	Dev	UAT	Production
Separate PostgreSQL instance/volume	Yes	Yes	Yes
Separate MinIO bucket/volume	Yes	Yes	Yes
Separate `JWT_SECRET_KEY`	Yes	Yes	Yes
`ALLOW_DEMO_LOGIN`	May be true	false	**false**
`ENABLE_API_DOCS`	May be true	false / limited	**false**
`CLAMAV_ENABLED`	Optional	Recommended	**true** (target)
VPN required for access	Optional	Yes	Yes (remote)
Direct DB port exposed to LAN	Local only	Restricted	**Admin only**

4. Network access model

User type	Dev	UAT	Prod
Engineer (office LAN)	Direct	HTTPS internal URL	Read-only tools — **Template / Pending sign-off**

Engineer (remote)	VPN + dev host	VPN + UAT URL	VPN + prod URL (break-glass)
Business UAT tester	—	VPN + UAT	No standing prod access
IT admin	SSH to hosts	SSH	SSH + backup

Production application URLs are **not** exposed as public internet SaaS endpoints.

5. Data handling between environments

Flow	Policy	Evidence
Prod → UAT refresh	Mask PII; approval required	**Template / Pending sign-off**
Prod → Dev	**Discouraged**; use fixtures	**Template / Pending sign-off**
UAT → Prod	Schema via Alembic only	Migration logs

6. Configuration management

Item	Storage	Notes
`.env` per environment	On respective hosts only	Not in Git
Docker Compose files	Git (`docker-compose.*.yml`)	Same files; env differs
TLS certificates	Per-host `docker/nginx/certs/`	Internal CA

7. Privileged access to production

Access type	Control	Logged
SSH to Docker host	Key-based; IT jump	**Template / Pending sign-off**
`docker exec` into API	Platform on-call only	**Template / Pending sign-off**
Database superuser	Break-glass account	**Template / Pending sign-off**
MinIO admin console	LAN/VPN + strong password	**Template / Pending sign-off**

8. Related documents

Ref	Document
1.3	Application Network Diagram
2.4	Change and Release Records
1.7	Infrastructure Shared-Responsibility Matrix

Approval: Template / Pending sign-off